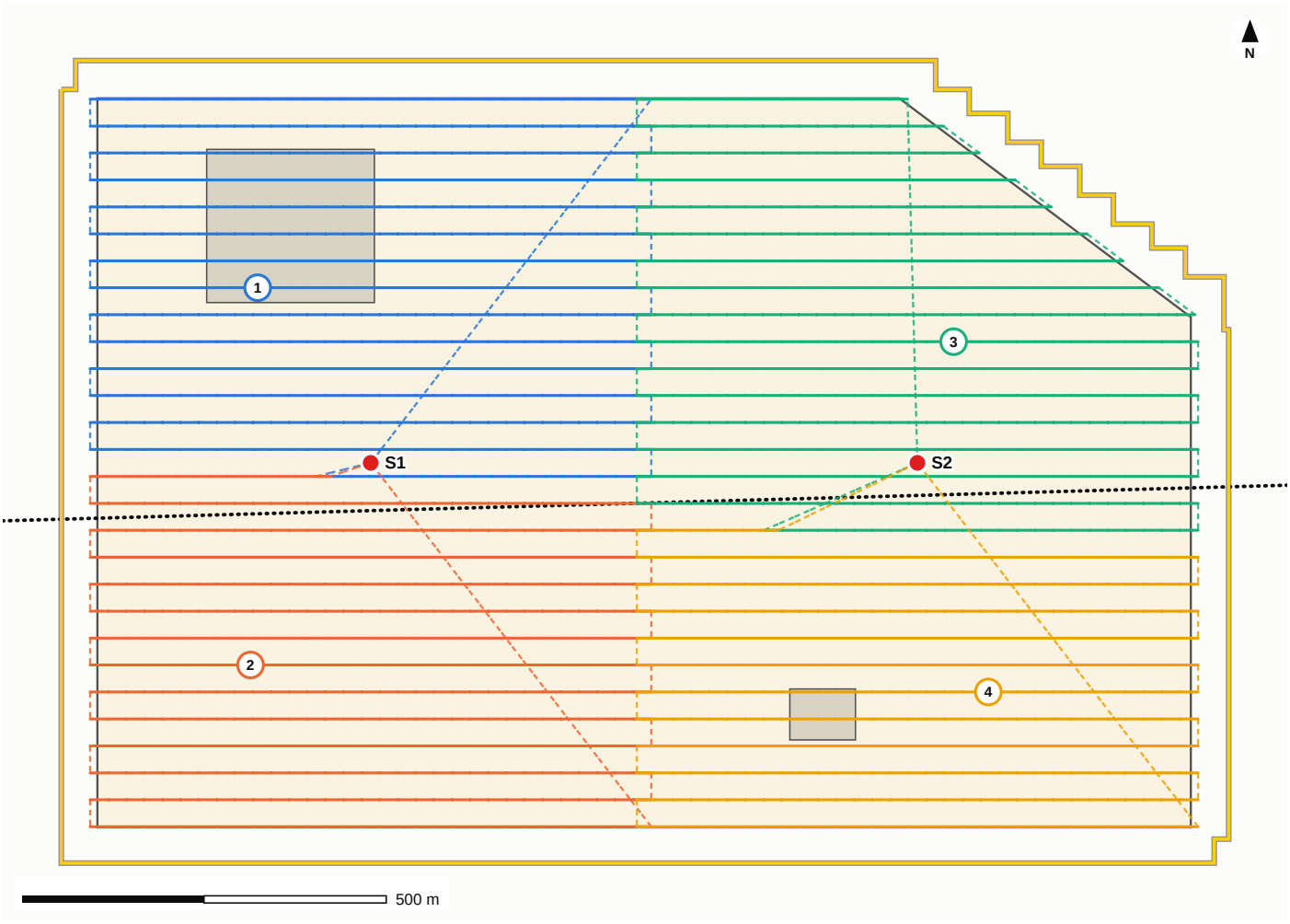
Lot 12 Stockpile & Paddock Survey

Drone mission field brief · DJI Matrice 4E · flight date Thu 15 Oct 2026 · prepared 13 Oct 2026

Summary

Flight date **Thu 15 Oct 2026 (UTC+8)**	Aircraft **DJI Matrice 4E**	Mission **area**	Area **144.00 ha**
Height **75 m**	Return-to-home height **60 m**	Battery-change height **60 m**	GSD **2.00 cm/px**
Speed **10.0 m/s**	Photos **2100**	Image coverage **169.25 ha**	Flights **4**
Flight time **1h 24m**	Usable / battery **25.4 min**	Wind **12 kn (6.0 m/s) from 220°**	Temp / payload **31 °C / 0.20 kg**
Accessory power **18 W**	Daylight **05:37 – 18:29**	Civil twilight **05:13 – 18:54**	Best light (sun >= 30°) **08:04 – 16:02**
On site **07:30 – 09:36 (2.1 h)**	Battery sets **4 (recommended 4)**	Charge waiting **0 min**	

Mission map



Flight 1 Flight 2 Flight 3 Flight 4 Transit to / from launch station Take-off / launch station
Image coverage (footprint extent) Obstacle
Grid north. Map and plan in the mission's planning coordinate system.

Payload

Item	Weight	Power
RTK module	0.11 kg	3 W
Spotlight	0.09 kg	15 W
Total	**0.20 kg**	**18 W**

Take-off mass 1.42 kg (limit 0.20 kg payload). Usable endurance 25.4 min per battery; 28.3 min without accessory power.

Launch stations

Station	Lat, Lon (WGS84)	Flights launched
S1	-32.292007, 115.740150	2
S2	-32.292086, 115.748114	2

Flights & battery plan

#	Launch -> land	Set	Start	End	Min	Battery	Photos	KP (km)	Sun	Notes	Done
1	S1 -> S1	1	07:45	08:08	22.6	99%	570		26–31° low sun	battery change mid-line.	☐
2	S1 -> S1	2	08:12	08:33	20.9	91%	524		32–36°		☐
3	S2 -> S2	3	08:46	09:09	22.8	100%	567		39–43°	battery change mid-line. drive 10 min	☐
4	S2 -> S2	4	09:13	09:31	18.1	79%	439		44–48°		☐

Terrain (DEM)

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Terrain & obstacle check:
  Lowest clearance  : 47 m  (limit 44 m)
  Highest above gnd : 75 m  (limit 120 m)
  GSD over terrain  : 2.00 - 2.00 cm/px (mean 2.00)
  Height over survey: 75 m - 75 m above ground
  Worst overlap     : 75% front / 65% side (where ground is highest)
  OK - no clearance or height-limit problems found.
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Warnings & checks

- Obstacles: 3 feature(s), tallest 28 m; the aircraft keeps 10 m above them within 15 m horizontally.
- LIGHT: flight 1 with the sun below 30° - long shadows reduce mapping quality. Best light 08:04–16:02 (sun >= 30°).
- Check airspace, NOTAMs, weather and the mission preview in the flight app before each flight.